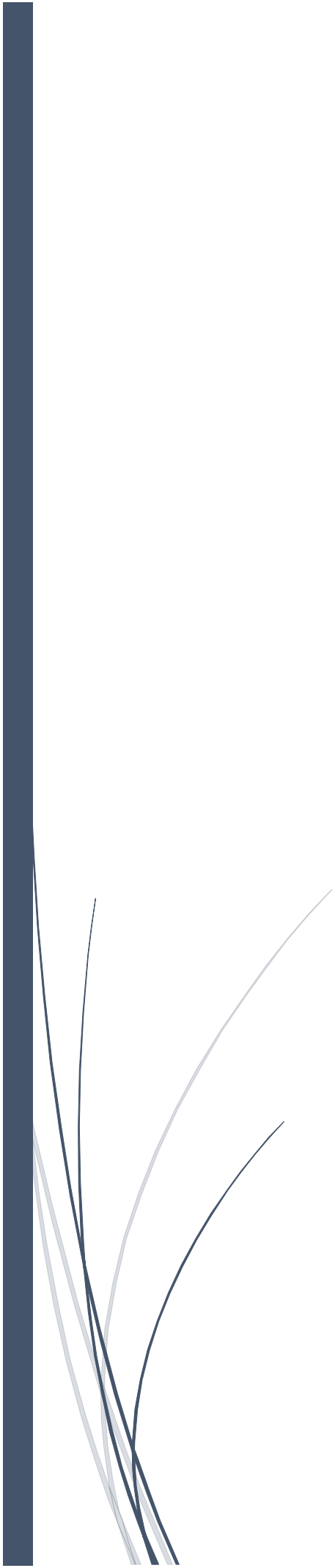


A thick dark blue vertical bar is positioned on the left side of the slide. From the bottom of this bar, several thin, curved lines in dark blue and light grey extend upwards and outwards, creating an abstract, organic shape.

SWE7303 - DevOps

Lab Title: DevOps Integration Lab

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Lab Objective:**In this lab, you will learn how to:**

1. Launch a basic web application on AWS using an EC2 instance.
2. Set up Git version control in a team development environment.
3. Deploy a web application using Docker containers.

Lab Environment:

- AWS account
- Git installed on your local machine.
- Docker installed on your local machine.

Lab Steps:**Step 1: Launch a Basic Web Application on AWS Using EC2 Instance****1. Sign up for an AWS Account:**

- Go to the AWS website and sign up for an account.
- Follow the instructions to create your account.

2. Launch an EC2 Instance:

- Launch a new EC2 instance with Amazon Linux AML.
- Connect to the EC2 instance using Direct connect or SSH.

3. Install a Web Server:

- Install Apache web server on the EC2 instance.
- Start the Apache service.

4. Create and Upload a Web Page:

- Create a basic HTML web page on the EC2 instance.
- Upload the web page to the EC2 instance.

5. Access your Web Application:

- Access the web application using the public IP address of the EC2 instance.

Step 2: Git Version Control in a Team Development Environment

1. Repository Setup:

- Create a new directory named "MyProject" on your local machine.
- Initialise a new Git repository in the "MyProject" directory.
- Add a sample file named "index.html" to the repository.
- Commit the initial file to the repository.

2. Branching and Merging:

- Create a new branch named "feature-login" for implementing the login feature.
- Modify the "index.html" file to include a login form.
- Commit the changes to the "feature-login" branch.
- Create another branch named "feature-payment" for implementing the payment feature.
- Modify the "index.html" file to include a payment form.
- Commit the changes to the "feature-payment" branch.
- Merge the "feature-login" branch into the "master" branch.
- Resolve any conflicts that may arise during the merge.

3. Collaboration and Conflict Resolution:

- Create a remote repository on GitHub.
- Push your local repository to the remote repository.
- Invite a team member to collaborate on the project.
- Ask your team member to clone the remote repository to their local machine.
- Your team member should create a new branch named "bug-fix" and fix a bug in the "index.html" file.
- They should commit the bug fix to the "bug-fix" branch and push it to the remote repository.
- Pull the latest changes from the remote repository to your local repository.
- Merge the "bug-fix" branch into the "master" branch.
- Resolve any conflicts that may arise during the merge.

Step 3: Deploying a Web Application with Docker

1. Set up the Project Directory:

- Create a new directory named "MyDockerWebApp" on your local machine.
- Copy your web application codebase (HTML, CSS, and JavaScript files) into this directory.

2. Create a Dockerfile:

- Create a file named "Dockerfile" in the project directory.
- Add Dockerfile instructions to package your web application into a Docker container.

3. Build the Docker Image:

- Open a terminal and navigate to the project directory.
- Run the Docker build command to build the Docker image.

4. Run the Docker Container:

- Once the Docker image is built, run the Docker container using the appropriate Docker run command.

5. Test the Web Application:

- Access the web application in a web browser using the exposed port of the Docker container.

Conclusion:

By completing this lab, you have gained hands-on experience in launching a web application on AWS using EC2, setting up Git version control for team development, and deploying a web application using Docker containers. These skills are essential for DevOps practitioners to streamline software development, collaboration, and deployment processes.